

FUNCTIONAL DOCUMENT

Smart Inventory Management System

Functional requirements, user roles, workflows, business rules, inputs, outputs, and acceptance criteria for the project.

Project

Smart Inventory Management System

Document Type

Functional Specification

Backend

Node.js and Express.js

Database

MySQL 8+

Frontend

HTML5, CSS3, Vanilla JavaScript

Prepared For

Project Submission

1. Project Purpose

The Smart Inventory Management System is designed to help a retail or warehouse business manage inventory, suppliers, customers, employees, sales orders, invoices, stock alerts, and business reports from a single web application.

The system reduces manual stock tracking by providing automated FIFO stock deduction, batch-level inventory control, role-based access, customer history, supplier intelligence, and real-time dashboard monitoring.

2. Scope of the System

- User login, logout, password reset, and password change
- Role-based access for admin, manager, and staff
- Product inventory creation, update, delete, approval, and search
- FIFO batch tracking and restocking
- Supplier management and supplier-product mapping
- Order creation, update, cancellation, deletion, and invoice generation
- Customer contact management and order history
- Employee/user management
- Dashboard analytics and stock alerts
- Export reports, audit logs, and database backup
- Email delivery for customer invoices and stock alerts

3. User Roles

Role	Main Responsibilities
Admin	Full system control, user management, product approvals, delete permissions, audit exports, and backup generation
Manager	Inventory oversight, supplier management, restocking, order updates, monitoring, and approvals workflow support
Staff	Daily operational work such as viewing stock, creating orders, and updating assigned records

4. Functional Requirements

4.1 Authentication and Account Access

The system shall allow users to log in using valid credentials. Passwords are stored securely using hashing. After successful login, the backend returns a JWT token used for protected API access.

- User can log in with username and password.
- User can log out from the application.
- Logged-in user details can be fetched from the current session.
- User can change password after authentication.
- Password reset is available from the login screen.
- Invalid or expired tokens are rejected.

4.2 Dashboard and Analytics

- Display revenue, profit, order, and inventory snapshots.
- Show stock alerts for low-stock and expiring items.
- Show pending product approvals for admin users.
- Display sales trends and category distribution using charts.
- Provide date-based sales report filtering.
- Allow report export for business use.

4.3 Inventory Management

- View all approved inventory items.
- Search and filter products.
- Add new product with SKU, category, price, supplier, expiry date, description, and low-stock point.
- Update and delete product records according to permissions.
- Calculate stock status as in-stock, low-stock, critical, or out-of-stock.
- Track inventory movements for restock, adjustment, sale, and return events.
- Show alternative suppliers for a product.

4.4 Product Approval Workflow

- Products created by admin can be approved immediately.
- Products created by non-admin users can be marked as pending.
- Admin can view, approve, or reject pending products.

4.5 FIFO Batch Management

- Each restock creates or updates inventory batch records.

- Batches store supplier, purchase cost, selling price, initial stock, current stock, received date, and expiry date.
- Admin and manager can view and update product batches.
- Orders consume stock from the oldest available batch first.
- Batch usage is stored for every order.
- Cancelled or deleted orders restore stock back to the original consumed batches.

4.6 Supplier Management

- View, add, update, and delete suppliers according to permissions.
- Store company name, contact details, status, lead time, delivery rate, quality rating, and payment terms.
- View products supplied by a supplier.
- View supplier intelligence and performance indicators.
- Link suppliers to inventory items with pricing, quality, lead-time, minimum order quantity, and preferred supplier status.

4.7 Order Management

- View, create, update, cancel, and delete orders according to permissions.
- Capture customer phone and email during order creation.
- Automatically create or update customer records when an order is placed.
- Deduct inventory using FIFO batch logic.
- Calculate order amount, cost, and profit based on consumed batches.
- View order batch breakdown for admin users.

4.8 Customer Management

- View, add, update, and delete customer details according to permissions.
- View all orders placed by a selected customer.
- Auto-fill customer name suggestions during order creation.

4.9 Employee Management

- Admin can view, add, update, and delete employee/user records.
- Employee records include role, status, department, and manager assignment.
- Role controls decide which screens and actions are visible to each user.

4.10 Invoice Management

- Generate invoice for completed orders.
- Display invoice with product, customer, order, price, total, and status details.
- Print invoice from the browser.

- Send invoice to customer email.
- Send admin copy/BCC when configured.
- Prevent invoice generation for invalid or unsupported order states.

4.11 Reporting and Exports

- Export daily order reports, sales reports, product reports, and staff reports.
- Export audit logs for admin users.
- Generate database backup for admin users.

4.12 Monitoring and Audit

- Show monitoring overview, recent audit logs, team health, critical alerts, inventory health, and category performance.
- Allow admin users to export audit logs and download database backup.

5. Major User Workflows

Login Workflow

User opens the application, enters credentials, receives a JWT token, and sees role-based screens.

Product Restock Workflow

Admin or manager selects a product, enters batch details, and the system updates stock and logs movement.

Sales Order Workflow

User selects customer and product, the system checks stock, deducts FIFO batches, saves order, and updates customer history.

Order Cancellation Workflow

System restores consumed quantities to original batches and recalculates product stock/status.

Invoice Workflow

User generates a printable invoice for a completed order and can send it to the customer by email.

6. Input and Output Summary

Module	Main Inputs	Main Outputs
Authentication	Username, password, reset/change password details	JWT token, user profile, success/error messages
Inventory	Product data, stock, price, supplier, expiry, low-stock point	Product list, stock status, alerts, movement logs
Batches	Quantity, supplier, purchase cost, selling price, expiry	FIFO batches, updated stock, batch usage
Suppliers	Company and performance details	Supplier list, intelligence, product mappings
Orders	Customer, product, quantity, status	Order records, invoice, stock deduction, profit calculation
Customers	Name, phone, email, address	Customer list and order history
Employees	User details, role, status, department	Employee records and access control
Reports	Date range, report type, export format	Reports, audit exports, backup files

7. Business Rules

- Only authenticated users can access protected application features.
- Admin users have the highest permission level.
- Product, supplier, customer, and employee deletion are restricted by role.
- Restocking is available to admin and manager users.
- FIFO inventory deduction must use oldest available batches first.
- Cancelled orders must restore stock to the same batches that were used.
- Product status must reflect current stock and low-stock rules.
- Completed order invoices can be generated and emailed.
- Audit and backup exports are restricted to admin users.

8. Non-Functional Requirements

Requirement	Description
Security	JWT authentication, bcrypt password hashing, and role-based authorization
Usability	Single-page interface with dashboard, tables, forms, modals, alerts, and charts
Reliability	Stock synchronization, FIFO restoration, and auto-bootstrapping database logic
Maintainability	Separate route modules, frontend modules, and shared helper libraries
Performance	MySQL connection pooling and focused API endpoints
Auditability	Activity logs, inventory movements, and audit export support
Availability	Local deployment through Node.js and MySQL with backup support

9. Assumptions and Constraints

- The application is intended for a local or small business deployment.
- MySQL must be running before the application starts.
- Valid environment variables must be configured in `.env`.
- Email invoice delivery requires SMTP configuration.
- The application currently handles sales orders around a selected inventory item.
- Admin users are responsible for backup, audit, and destructive operations.

10. Acceptance Criteria

The system is acceptable when users can log in, see role-appropriate screens, manage inventory/suppliers/customers/orders, restock products with FIFO batches, restore stock after cancellation, generate invoices, view dashboard analytics, and export authorized reports or backups successfully.